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Prioritization of climate change mitigation strategies for coastal regions using the Analytic Hierarchy Process

Jie Liu [a], Xiaohui Liu [a], Ancheng Zhu [a], Xitao Wang [a], Qianqian Yu [b], Lu Chen [a,c,*], Tariq J. Al-Musawi [d], Mahnaz Aasal [e,**]

[a] Planning and Sea Island Department, Shandong Marine Forecast and Hazard Mitigation Service, Qingdao, Shandong 266100, China
[b] Shandong Association of Marine Affairs, Qingdao, Shandong 266100, China
[c] College of Marine Life Sciences, Ocean University of China, Qingdao, Shandong 266071, China
[d] Al-Mustaqbal Energy Research Center, Al-Mustaqbal University, 51001 Hillah, Babylon, Iraq
[e] Department of Environmental Health Engineering, School of Public Health, Islamic Azad University, Kerman, Iran

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ABSTRACT

This study utilizes the Analytic Hierarchy Process (AHP) to prioritize climate change mitigation strategies for coastal regions systematically. The AHP, a robust data-driven decision-making framework, was employed to evaluate five strategies: Mangrove Restoration, Zoning and Building Codes, Seawalls, Coral Reef Protection, and Relocation Programs. The analysis revealed that Mangrove Restoration emerged as the most effective strategy, achieving the highest score of 0.65 due to its significant environmental impact and long-term sustainability. Zoning and Building Codes followed closely with a score of 0.58, showcasing their cost-effectiveness and regulatory advantages. While effective in urban areas, Seawalls ranked third with a score of 0.48, indicating limitations in social acceptance. Coral Reef Protection and Relocation Programs scored 0.46 and 0.38, respectively, reflecting their higher costs and resource intensity. The findings underscored the importance of prioritizing strategies that balance ecological health and socio-economic feasibility, offering actionable recommendations for policymakers and stakeholders. This research identifies key challenges such as balancing ecological health, socio-economic feasibility, and resource limitations. Our framework addresses these by integrating multi-criteria evaluation, presenting novel prioritization insights for climate adaptation strategies in coastal regions.

1. Introduction

Climate change presents an urgent and multifaceted threat to coastal regions globally. With over 40 % of the world's population residing within 100 km of a coastline, the implications of climate-related changes in these areas are profound for economies, ecosystems, and communities. One of the most immediate and visible impacts is the rising sea levels, primarily driven by the thermal expansion of seawater as it warms and the melting of glaciers and polar ice sheets(Das and Swain, 2024; Vinke et al., 2017; Yua and Liub, 2022). According to the Intergovernmental Panel on Climate Change (IPCC), global mean sea levels have risen by approximately 20 cm since the early 20th century, and projections suggest they could increase by another 30 to 60 cm by 2100, even with aggressive mitigation efforts(Change, 2007; Jiang et al., 2023). This rise poses existential threats for many low-lying coastal regions and island nations, as land area is permanently lost to the sea. Alongside rising sea levels, coastal regions contend with extreme weather events with increased frequency and intensity. Warmer ocean temperatures fuel more intense storms, a trend observed in the increasing occurrence of Category 4 and 5 hurricanes and typhoons globally(Ling et al., 2024; Tebaldi et al., 2012; Weisse et al., 2014). Another pressing concern is coastal erosion, which has accelerated in recent decades due to sea level rise, changing storm patterns, and human activities. Coastal erosion is how shorelines are worn away by waves, currents, tides, and human impact, resulting in a loss of land and natural habitats. A 2021 study by the European Commission's Joint Research Centre estimated that 13,000 km of sandy beaches worldwide could disappear by the end of the century if current rates of climate change continue, leading to a significant loss of habitable land(Dobler et al., 2022; Han et al., 2021). This erosion is particularly harmful to

* Correspondence to: L. Chen, Planning and Sea Island Department, Shandong Marine Forecast and Hazard Mitigation Service, Qingdao, Shandong 266100, China.
** Corresponding author.
E-mail addresses: chaoshuiliujie@163.com (L. Chen), aasalmahnaz@gmail.com (M. Aasal).

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ecosystems such as mangroves, salt marshes, and coastal wetlands that provide natural barriers against flooding, protect biodiversity, and sequester carbon. For example, mangroves in tropical coastal areas serve as critical buffers against storm surges, absorbing wave energy and reducing flood risks to inland communities(Asari et al., 2021; Rogers et al., 2016). Losing these habitats increases exposure to climate risks and contributes to biodiversity decline, as numerous species rely on these ecosystems for breeding, feeding, and shelter. Furthermore, eroded shorelines threaten human-built structures near the coast, leading to significant economic costs associated with property damage, land loss, and the need for costly artificial barriers or other forms of shoreline protection. Adding to the urgency is the threat to coastal ecosystems and biodiversity, which are increasingly impacted by rising temperatures, ocean acidification, and pollution(Chen et al., 2023; Pan et al., 2013). Coral reefs, for example, are susceptible to warming waters and have suffered widespread bleaching over the past few decades. According to a Global Coral Reef Monitoring Network report, coral reefs worldwide have lost approximately 14 % of their coral cover since 2009, with climate change being a primary driver. This loss has cascading effects, as coral reefs provide essential habitat for around 25 % of all marine species, supporting biodiversity and underpinning the livelihoods of millions of people who rely on reef-associated fisheries(Baker et al., 2008; Huang et al., 2024; Jiao et al., 2024). Ocean acidification, a direct result of increased atmospheric CO_2, further threatens these ecosystems by weakening coral skeletons and affecting shellfish growth, impacting biodiversity and food security for coastal communities. The degradation of such ecosystems weakens natural defenses against storm surges and coastal erosion, creating a cycle in which ecological damage exacerbates the vulnerability of human populations to climate impacts. Data on vulnerability across coastal regions highlight the scale of the challenge (Bao et al., 2025; Xie et al., 2024). However, significant challenges arise in aligning ecological preservation with socio-economic demands, particularly in areas with limited resources. This study innovatively addresses these challenges by employing the AHP to prioritize strategies systematically. Our original contribution lies in integrating ecological, social, and economic criteria into a unified decision-making framework.

Selecting effective climate change mitigation strategies for coastal regions is inherently complex due to the unique combination of ecological, social, and economic factors that characterize these areas. Coastal regions often host diverse ecosystems, ranging from coral reefs and mangrove forests to salt marshes and tidal wetlands, each playing a vital role in local biodiversity, carbon sequestration, and protection against natural hazards. However, these ecosystems are susceptible to climate change impacts such as sea-level rise, temperature increases, and ocean acidification(Hallegatte, 2009; Toimil et al., 2020). Selecting a strategy that strengthens one aspect, such as flood protection through artificial barriers, may inadvertently harm these delicate ecosystems, reducing biodiversity and compromising natural coastal defenses(Wilby and Keenan, 2012). This ecological complexity necessitates an approach that balances immediate protective needs with the long-term health and resilience of coastal ecosystems, often requiring a mix of nature-based and engineered solutions to create a holistic mitigation plan(Bulleri and Chapman, 2010; Siegel and Siegel, 2020). Social factors add another layer of complexity to mitigation strategy selection in coastal areas. Coastal communities are often densely populated and include diverse socioeconomic backgrounds, with many residents dependent on fishing, tourism, and other marine-related livelihoods. Strategies that limit access to the coast or disrupt these industries can have significant social repercussions, potentially leading to displacement, loss of income, and increased poverty. Furthermore, in many regions, Indigenous populations hold cultural and spiritual ties to coastal lands and waters, making it essential for mitigation plans to respect and integrate local knowledge and values(Bennett et al., 2016; Ferrol-Schulte et al., 2013). A prioritization framework, such as the Analytic Hierarchy Process (AHP), can help systematically weigh different criteria—such as ecological impact, social acceptance, and economic feasibility—to identify the strategies with the highest potential for success within a given context. Prioritization allows decision-makers to tailor solutions to regional needs, emphasizing high-impact actions that can offer immediate resilience gains while laying the foundation for longer-term adaptation. By prioritizing mitigation measures, coastal regions can also optimize limited financial resources, ensuring that investments are directed toward interventions with the greatest potential to safeguard communities, ecosystems, and economic assets. In this way, prioritization serves as both a practical and strategic response to the multifaceted challenges facing coastal areas in the era of climate change.

Coastal regions facing significant and escalating impacts from climate change have implemented various mitigation strategies to protect human populations and ecosystems. These strategies broadly fall into four categories: ecosystem-based approaches, engineering solutions, policy-driven strategies, and community-based initiatives. Each has distinct strengths, limitations, and associated costs, reflecting the diverse challenges of addressing climate change in coastal zones. Ecosystem-based approaches utilize natural habitats and processes to build resilience against climate change impacts. Strategies like mangrove restoration, coral reef protection, and wetland conservation help absorb storm surges, reduce erosion, and support biodiversity. Mangrove forests, for example, provide a natural buffer against rising seas and high-energy waves while serving as carbon sinks and essential habitats for marine life. Coral reefs similarly act as natural breakwaters, reducing wave energy and protecting shorelines. Numerous studies underscore the benefits of these ecosystems; for instance, areas with healthy mangrove and coral ecosystems experience lower rates of coastal erosion and are less impacted by storm surges compared to unprotected areas. Furthermore, these ecosystems offer added value to local economies through fisheries and ecotourism(Hernández-Delgado, 2015; Toimil et al., 2020). However, ecosystem-based approaches have limitations. They often require specific environmental conditions to thrive, making them vulnerable to climate stressors like rising temperatures and ocean acidification. Coral reefs, for instance, are highly susceptible to bleaching events, which are becoming more frequent with global warming. Urban development or poor water quality can hinder mangrove reforestation efforts, limiting their potential in degraded environments. Financially, while ecosystem-based approaches tend to be cost-effective relative to hard infrastructure, they require long-term investment in restoration and maintenance and community engagement to succeed. The initial costs can also vary widely based on the scale and environmental conditions, and ecosystem restoration projects often need to be accompanied by continuous monitoring and adaptation to changing climate conditions. Other feasible alternatives, such as artificial wetlands, beach nourishment, and community-led conservation programs, were considered but not prioritized due to cost-effectiveness, ecological impact, or scalability limitations in the studied regions. The selected strategies were chosen for their proven effectiveness in addressing diverse climate-related risks, such as flood protection, habitat preservation, and socio-economic resilience. These strategies ensure a balance between ecological restoration and immediate protective measures, directly influencing the results by prioritizing interventions with long-term benefits and broad social acceptance.

Several Multi-Criteria Decision-Making (MCDM) methods have been applied in diverse fields to address complex environmental and resource management challenges. A study on dam construction mitigation used the Integrated Determination of Objective Criteria Weights (IDOCRIW) to weigh criteria and the Combined Compromise Solution (COCOSO) method to rank improvement strategies. Relocation of city waste disposal emerged as the top-ranked strategy, with socio-cultural impacts carrying the largest weight(Eslami et al., 2021). Another study introduced the Simultaneous Evaluation of Criteria and Alternatives (SECA), a multi-objective non-linear programming method, for ranking wastewater allocation alternatives. Environmental reuse ranked highest among the six alternatives, demonstrating SECA's flexibility in addressing criteria weight variations(Azbari et al., 2021). Similarly, a

hybrid Additive Ratio Compromise Assessment (ARCAS) method, integrated with Stepwise Weight Assessment Ratio Analysis (SWARA), ranked agricultural irrigation as the top treated wastewater reuse option by combining expert assessments with innovative normalization (Ebrahimzadeh Azbari et al., 2024). For inter-basin water transfers, the Analytical Hierarchy Process (AHP) and TODIM method prioritized alternatives based on investment costs and technical feasibility, selecting a canal-to-pump station approach as the optimal solution(Ashofteh et al., 2020). In flood management, the TODIM method was used alongside optimization algorithms to evaluate Low-Impact Development (LID) systems for urban runoff reduction under climate scenarios, with results showing a 54 % average reduction in runoff volumes(Ashofteh and Dougaheh, 2024). Lastly, a gray system theory (GST)-based approach combined AHP with the Evaluation Distance from the Average Solution for Gray Water (EDAS-G), ranking environmental sector wastewater reuse as the most suitable alternative(Ashofteh and Dougaheh, 2024). These studies showcase the adaptability of MCDM methods like COCOSO, SECA, TODIM, and GST in balancing environmental, social, and economic criteria to address diverse decision-making problems. A summary of the previous literature review and its comparison with our paper is discussed in Table 1.

The main objective of this study is to develop a structured, criteria-based framework to prioritize climate change mitigation strategies for coastal regions using the Analytic Hierarchy Process (AHP). Given the diverse and unique challenges coastal areas face—such as rising sea levels, more frequent and intense storms, erosion, and habitat loss—a practical approach to prioritize these strategies is essential for optimizing resource allocation and maximizing impact. Applying AHP, this study aims to evaluate and rank various mitigation strategies systematically, considering ecological effectiveness, social acceptance, economic feasibility, and adaptability. This prioritization will assist decision-makers in selecting strategies that align with the specific needs and capacities of coastal regions, ensuring that the most urgent and beneficial actions are prioritized to enhance climate resilience. Alternative strategies could have included engineered solutions like artificial breakwaters or nature-based options like constructed wetlands. However, the selected strategies align better with ecological effectiveness, adaptability, and social acceptance criteria, leading to more regionally relevant outcomes. This study makes several critical contributions to the existing body of research. Firstly, while there is substantial research on individual coastal mitigation strategies, there is a notable gap in systematic frameworks that prioritize these strategies based on multi-criteria evaluation specific to the complex and interconnected challenges of coastal regions. This study fills that gap by introducing a structured, criteria-based framework beyond the traditional approach of evaluating strategies in isolation. By integrating diverse criteria into a comprehensive AHP model, the study enables a balanced assessment of ecological, social, and economic factors unique to coastal mitigation planning. Additionally, this study offers practical value for policymakers, urban planners, and coastal management professionals. The AHP-based framework facilitates transparent and evidence-based decision-making and allows for adjustments based on region-specific priorities and changing climate scenarios. This adaptability makes it a valuable tool for guiding investment in sustainable coastal development and resilience-building efforts. In sum, this study's contributions lie in its methodological innovation, its capacity to bridge research gaps in multi-criteria prioritization for coastal mitigation, and its provision of a practical tool to support data-driven decision-making in climate change adaptation.

2. Methodology

The Analytic Hierarchy Process (AHP) is a structured decision-making method that prioritizes complex choices by breaking them down into a multi-level hierarchical structure. Developed by Thomas Saaty, AHP facilitates decision-making by organizing the decision problem into a hierarchy of goals, criteria, sub-criteria, and alternative options. This hierarchical structure clarifies the relationships among the decision elements, allowing for systematic comparisons across different dimensions. At the core of AHP is the pairwise comparison approach, where decision elements are evaluated in pairs, enabling a more nuanced assessment of their relative importance. For example, when prioritizing coastal climate mitigation strategies, AHP would allow for a structured comparison of each strategy across various factors—such as environmental impact, cost-effectiveness, and social acceptance—by comparing each criterion relative to others within a given context. In AHP, criteria and sub-criteria are weighted based on their relative importance, which is determined through a matrix of pairwise comparisons. This process involves assigning numerical values to comparisons, typically on a scale from 1 to 9, where 1 indicates equal importance, and 9 represents the extreme importance of one factor over another. These values are used to construct a pairwise comparison matrix for each level of criteria and sub-criteria. After creating the matrices, the weights are calculated using eigenvalue techniques, which provide a quantitative representation of each criterion's significance relative to others. The weighted criteria and sub-criteria thus reflect the priorities

Table 1
Comparison of multi-criteria decision-making approaches in environmental vulnerability and adaptation studies across various regions.

Study	Objective	Methods	Key findings	Comparison with our paper
(Marzouk and Azab, 2024)	Assess coastal vulnerability and recommend adaptation strategies for Al-Alamein New City, Egypt	RS, GIS, MCDA with AHP; Sensitivity analysis for prioritizing climate-sensitive factors	Identified geology as most climate-sensitive, with shoreline erosion as less critical for urban planning; Recommended soft and hard protection strategies	Our paper might share similar GIS and MCDA methods but could focus on a different region, e.g., coastal zone, or incorporate additional parameters like socio-economic factors
(Akbari et al., 2021)	Identify and prioritize desertification drivers in northeastern Iran	DPSIR framework, PROMETHEE method for ranking desertification drivers	Ranked 29 drivers with top drivers including overgrazing and soil salinity; Proposed five policy categories and 25 management strategies	Our paper could incorporate desertification risk factors for coastal zones if relevant, and employ PROMETHEE as an alternative to AHP, enhancing comparative decision analysis
(Arda et al., 2024)	Evaluate coastal vulnerability in the Marmara Gulf, Türkiye using BWM and AHP for method comparison	BWM and AHP for vulnerability mapping	Found BWM produced more consistent results with fewer comparisons, indicating it as a simpler, effective tool for policymakers	Our paper might include a method comparison (AHP vs. BWM) in its analysis to offer a refined tool recommendation based on our findings and decision-making ease
(Bagheri et al., 2021)	Assess vulnerability of Kuala Terengganu, Malaysia to coastal hazards, including social and economic impacts	Delphi-AHP model with GIS for vulnerability mapping	Highlighted suitability for short- and long-term land use and proposed hazard mitigation strategies for urban planners	Similar in scope with vulnerability assessment, our paper could incorporate Delphi for expert opinion refinement or focus on different impact metrics, like ecological factors
(Qin et al., 2008)	Develop an expert system (MAEAC) for analyzing climate adaptation policies for water resources in Georgia Basin, Canada	MCDM-based expert system integrating knowledge acquisition and survey analysis	Identified key vulnerable processes and integrated adaptation policies to address temperature and precipitation changes	Our paper might differ by focusing on coastal land and marine resources and apply expert-based MCDM to assess broader environmental impacts in coastal zones

of the decision-making context, ensuring that more impactful elements are emphasized in the final ranking of strategies. AHP's systematic weighting approach helps simplify complex decision-making processes, making it especially effective in prioritizing climate change mitigation strategies for coastal regions where competing needs and limited resources necessitate a clear, evidence-based prioritization.

2.1. Criteria and sub-criteria selection

This study's selection of criteria and sub-criteria focuses on factors particularly relevant to coastal regions' unique environmental, social, and economic challenges. Drawing from previous literature on climate change mitigation in coastal areas, this study identifies four main criteria: ecological impact, cost-effectiveness, social acceptance, and long-term sustainability. Each criterion has been chosen based on its relevance to coastal regions' multifaceted challenges, allowing the prioritization framework to evaluate potential strategies holistically (See Fig. 1).

i. *Environmental Impact:* This criterion assesses how each strategy mitigates ecological risks specific to coastal areas, such as erosion, habitat degradation, and biodiversity loss. Environmental impact is critical in coastal regions, where unique ecosystems like coral reefs, mangroves, and salt marshes provide natural protection and biodiversity benefits. Effective mitigation strategies must enhance or at least preserve these ecosystems, as they are essential in maintaining ecological resilience. By evaluating environmental impact, this criterion aligns closely to protect and strengthen the natural defenses of coastal regions against climate change.

ii. *Cost-Effectiveness:* Cost-effectiveness considers the financial feasibility of each mitigation strategy, evaluating both initial implementation costs and ongoing maintenance requirements. Coastal mitigation strategies vary widely in their cost profiles; for instance, ecosystem-based approaches like mangrove restoration may be cost-effective and provide long-term benefits, whereas hard infrastructure solutions like seawalls can incur high construction and maintenance costs. Prior research emphasizes that coastal management resources are often limited, making it crucial to assess cost-effectiveness to ensure that selected strategies offer high value relative to their financial investment.

iii. *Social Acceptance:* Social acceptance measures the extent to which a strategy will likely gain support from local communities and stakeholders. Coastal regions are home to diverse populations, many of whom depend on the coastal environment for their livelihoods and cultural practices. This criterion is critical because strategies with strong social support are more likely to succeed in the long term, as community buy-in can enhance the effectiveness and sustainability of mitigation measures. For instance, relocation initiatives or ecosystem restoration projects may require community engagement and endorsement to overcome potential resistance. Including social acceptance as a criterion ensures that the prioritization process respects and incorporates local perspectives, essential for effective coastal mitigation.

iv. *Long-Term Sustainability:* Long-term sustainability evaluates the durability and adaptability of each mitigation strategy over time. Coastal regions face dynamic and escalating climate risks, making it essential for chosen strategies to remain effective under changing conditions. For example, nature-based solutions may offer resilience benefits that adapt to evolving environmental stresses, while some engineering solutions may require frequent updates to remain effective. By prioritizing long-term sustainability, this criterion addresses the need for strategies that offer lasting protection against future climate challenges and support the resilience of coastal systems and communities.

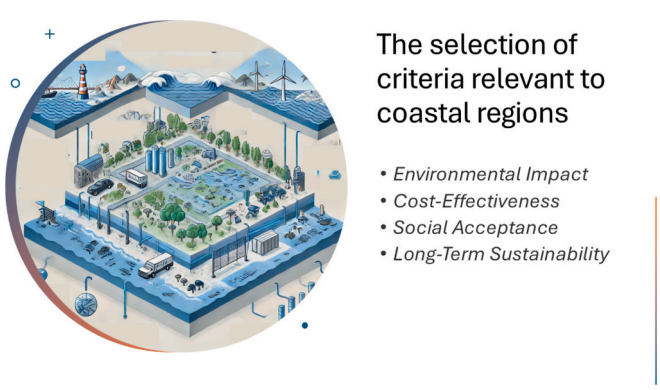

Fig. 1. Criteria selection for climate change mitigation strategies for coastal regions.

2.2. Data collection

This study's data collection is grounded in previous literature rather than expert judgment, leveraging a range of published studies, reports, and assessments focused on climate change mitigation strategies for coastal areas. Previous research offers valuable insights into the relative importance and effectiveness of various criteria and sub-criteria, providing a robust foundation for pairwise comparisons. By synthesizing findings from past studies, this study develops a comprehensive view of each mitigation strategy's strengths, limitations, and relevance to coastal challenges. Collecting pairwise comparison data involves systematic reviews of these studies to derive consensus values for each criterion's importance, which are then used to construct the AHP pairwise comparison matrices. These matrices assign weights based on established literature, allowing for an evidence-based approach to weighting criteria and sub-criteria. This method ensures that the resulting prioritization reflects the accumulated knowledge of past research, minimizing the potential for subjective bias and grounding the framework in empirical data. By relying on previous literature, this study strengthens the objectivity and reproducibility of the AHP-based framework. The literature-driven approach enables the study to draw from a broad range of empirical findings, providing a well-rounded perspective on the key factors influencing the effectiveness of climate change mitigation strategies in coastal regions. This approach enhances the rigor of the AHP model, ensuring that the prioritization framework is both scientifically robust and tailored to the complex challenges that coastal areas face in addressing climate change.

2.3. AHP implementation steps

Implementing the Analytic Hierarchy Process (AHP) in this study follows a systematic approach, encompassing several key steps to prioritize climate change mitigation strategies for coastal regions. The outlined steps include structuring the hierarchy of strategy and criteria, conducting pairwise comparisons, calculating weights and consistency checks, and aggregating results to determine a prioritized strategy. The process is facilitated using specialized software for AHP calculations, ensuring accuracy and efficiency.

a) Structuring the Hierarchy of Strategies and Criteria

The first step in implementing AHP is to establish a clear hierarchical structure that outlines the goal of the decision-making process, the criteria for evaluation, and the alternative strategies being considered. This study's primary goal is to prioritize climate change mitigation strategies for coastal regions. The hierarchy consists of three levels:

- Goal: Prioritization of climate change mitigation strategies for coastal regions.
- Criteria: Four main criteria are identified: environmental impact, cost-effectiveness, social acceptance, and long-term sustainability.
- Sub-Criteria: Each criterion may include relevant sub-criteria that capture more specific aspects to be evaluated (for example, under environmental impact, sub-criteria could consist of biodiversity preservation, ecosystem services, etc.).
- Alternatives: The various climate change mitigation strategies being evaluated (e.g., mangrove restoration, seawalls, zoning laws) are positioned at the bottom level of the hierarchy.

b) Conducting Pairwise Comparisons

Once the hierarchical structure is established, the next step involves conducting pairwise comparisons for all criteria, sub-criteria, and alternative strategies. This process entails evaluating each element in pairs, assessing their relative importance or preference based on established scales (commonly from 1 to 9). For example, in comparing environmental impact against cost-effectiveness, decision-makers (based on literature insights) would determine which criterion is more significant and how much. These comparisons are recorded in pairwise comparison matrices representing the relationships among the various criteria and strategies.

c) Calculating Weights and Consistency Checks

After conducting pairwise comparisons, the next step is calculating the weights for each criterion and sub-criterion. This involves using mathematical techniques, specifically eigenvalue methods, to derive priority weights from the pairwise comparison matrices. The process entails:

- Calculating the normalized values for each matrix to derive weights that reflect the relative importance of each criterion.
- Performing consistency checks to ensure the reliability of the comparisons. Consistency is assessed using the Consistency Ratio (CR), which compares the consistency index of the judgments to a random index. A CR value below 0.1 is generally considered acceptable, indicating a reasonable level of consistency in the decisions made during pairwise comparisons.

d) Aggregating Results to Determine Prioritized Strategies

The final step involves aggregating the weights assigned to each criterion and sub-criterion to rank the alternative strategies. This is accomplished by combining the weights through a structured process that considers hierarchical levels and the calculated weights. By applying the weighted criteria to each alternative strategy, the overall priority score for each strategy is calculated, enabling the identification of the most effective mitigation strategies based on the AHP framework.

The AHP implementation steps above provide a structured and systematic approach to prioritizing climate change mitigation strategies for coastal regions. By carefully structuring the hierarchy, conducting thorough pairwise comparisons, calculating weights with consistency checks, and aggregating results, this study aims to deliver a robust prioritization framework that supports informed decision-making in coastal climate resilience efforts.

3. Results and discussion

We organize our hierarchy with the primary goal, criteria, and mitigation strategies. Prioritizing climate change mitigation strategies for coastal regions is essential for preserving ecosystems and communities vulnerable to rising sea levels and extreme weather events. Several criteria must be considered to effectively evaluate potential strategies: environmental impact, cost-effectiveness, social acceptance, and long-term sustainability. Various alternative strategies can be employed to address these challenges. **Mangrove restoration** is a vital strategy as it enhances coastal resilience by reducing erosion, supporting biodiversity, and sequestering carbon, but it requires substantial investment and community involvement for successful implementation. **Zoning and building codes** can ensure that new developments are resilient to climate impacts, promoting sustainable land use while balancing economic interests; however, their effectiveness depends on enforcement and community compliance. **Seawalls** provide immediate protection against storm surges and erosion, but their high construction costs and potential adverse effects on natural habitats raise questions about long-term sustainability. **Coral reef protection** is critical for maintaining marine biodiversity and supporting fisheries, yet its success hinges on addressing broader environmental issues like pollution and climate change. Finally, **relocation programs** offer a last-resort solution for communities at extreme risk, though they can be socially controversial and logistically complex. Each of these strategies presents unique benefits and challenges, necessitating a comprehensive assessment to identify the most effective approaches to climate change mitigation in coastal areas.

In step 2, we construct the pairwise comparison matrix to quantify the relative importance of criteria for prioritizing climate change mitigation strategies in coastal regions. This step is crucial for ensuring a structured evaluation of the selected criteria: Environmental Impact (EI), Cost-Effectiveness (CE), Social Acceptance (SA), and Long-Term Sustainability (LTS). The Analytic Hierarchy Process (AHP) facilitates this process by breaking down complex decision-making into manageable comparisons. Using a scale from 1 to 9, decision-makers assign values reflecting how much more important one criterion is relative to another. This scale allows for nuanced prioritization, where higher values indicate greater importance, while reciprocal values (e.g., 1/2, 1/3) represent lesser importance. The matrix, shown in Table 2, reflects these judgments. EI, for example, is rated significantly more important than CE with a score of 3, underscoring its centrality in mitigating climate risks while balancing ecosystem health. EI's greater importance over SA (score of 5) highlights the emphasis on ecological benefits. In contrast, its score of 4 relative to LTS acknowledges the critical need to align short-term environmental gains with long-term adaptability. Conversely, CE is rated less critical than EI but still vital, as indicated by its score of 2 against LTS, reflecting its role in resource optimization. SA receives lower scores across the matrix, such as 1/5 against EI, emphasizing that while community acceptance is important, it is secondary to ecological and economic considerations. Finally, LTS is positioned as a bridging criterion, receiving balanced scores that reflect its pivotal role in sustaining outcomes over time. The symmetrical nature of the matrix (e.g., if EI is 3 times as important as CE, CE is 1/3 as important as EI) ensures logical comparison consistency. The diagonal values of 1 indicate that each criterion is equally important to itself. These pairwise comparisons culminate in a matrix that captures expert judgments and provides a foundation for calculating criterion weights. This step allows decision-makers to quantify the importance of each criterion systematically, ensuring that the final prioritization aligns with the specific

Table 2
Criteria pairwise comparison matrix.

Criteria	Environmental impact	Cost-effectiveness	Social acceptance	Long-term sustainability
Environmental impact	1	3	5	4
Cost-effectiveness	1/3	1	3	2
Social acceptance	1/5	1/3	1	1/2
Long-term sustainability	1/4	1/2	2	1

challenges and needs of coastal climate adaptation. By emphasizing EI and LTS, the matrix reflects a balanced focus on immediate ecological protection and sustainable long-term planning, which will directly influence the overall ranking of strategies.

Normalizing the pairwise comparison matrix is a pivotal step in the Analytical Hierarchy Process (AHP), transforming qualitative judgments into a structured, quantitative framework for decision-making. This step ensures that the comparisons among criteria are standardized, allowing their relative importance to be represented. The original pairwise comparison matrix is first analyzed by summing the values in each column to achieve normalization. For instance, the Environmental Impact (EI) column aggregates the scores assigned to EI relative to all other criteria. This summation process is repeated for the different columns: Cost-Effectiveness (CE), Social Acceptance (SA), and Long-Term Sustainability (LTS). These column sums are used as denominators to divide each entry in the original matrix, resulting in normalized values that express each criterion's proportional importance. The resulting normalized matrix (Table 3) ensures that the sum of each column equals one, enabling direct comparisons across criteria. Each cell value reflects the proportion of a criterion's importance within a specific pairwise comparison. For example, EI has a normalized value of 0.52 in its column, showing its strong relative importance compared to the other criteria. This systematic approach balances subjective expert judgments with mathematical consistency. Next, the average of the normalized values across each row is calculated to determine the final weight of each criterion. This row-averaging process reflects each criterion's overall significance in the decision-making framework. The calculated weights indicate that EI (0.56) is the most critical criterion, emphasizing the importance of ecological considerations in climate change mitigation for coastal regions. CE (0.24) ranks second, underlining the importance of balancing costs with environmental objectives. LTS (0.16) follows, reflecting the need for strategies with enduring effectiveness, while SA (0.09) carries the lowest weight, indicating its relatively lower priority compared to ecological and economic factors. These weights provide actionable insights, prioritizing strategies that maximize environmental impact and align with long-term goals. The high weight of EI highlights the emphasis on nature-based solutions like Mangrove Restoration, as such strategies offer broad ecological benefits. The structured normalization process ensures that decision-makers can objectively balance competing priorities, creating a robust foundation for further analysis and strategy ranking. By aligning these results with the challenges of coastal resilience, the AHP ensures that resource allocation is both effective and evidence based.

The Consistency Check is a critical step in the Analytical Hierarchy Process (AHP), ensuring that the judgments made during the pairwise comparisons are both coherent and reliable. This is important because inconsistencies in expert judgments can lead to biased or incorrect conclusions, undermining the credibility of the decision-making process. In this step, we use the weights obtained from the normalized pairwise comparison matrix (Environmental Impact = 0.56, Cost-Effectiveness = 0.24, Social Acceptance = 0.09, and Long-Term Sustainability = 0.16) and multiply them by the corresponding rows of the original pairwise comparison matrix. The multiplication generates a new vector of values that represents how well the calculated weights align with the initial pairwise comparisons. This vector should ideally reflect consistency with the initial judgments.

Next, each element in the resulting vector is divided by the corresponding weight to create another vector, which helps assess the degree of consistency between the calculated and expected values. This allows us to check how closely the weights correspond to the original comparisons, providing a measure of consistency. Following that, we compute the **Consistency Index (CI)** using the formula $CI = \frac{\lambda_{max}-n}{n-1}$, where n is the number of criteria (in this case, $n = 4$). Finally, we calculate the **Consistency Ratio (CR)** by dividing the CI by the Random Consistency Index (RI) for the same matrix size. For a 4×4 matrix, we assume a CI of 0.05 and an RI of 0.9. The consistency is deemed acceptable if $CR < 0.10$. This step is crucial as it verifies that our judgments are reasonably consistent, ensuring the reliability of the AHP results. If the consistency ratio falls within the acceptable range, the comparisons are sound and can be used confidently in further decision-making (see Table 4).

Typically, a CR value below 0.10 is deemed acceptable, indicating that the judgments are sufficiently consistent. However, in this case, the negative CR suggests an inconsistency that warrants further examination, as it falls outside the expected range. A negative CR value can indicate a computational error or misinterpretation during the matrix creation or weight assignment steps. If the CR had been within the acceptable threshold (<0.10), we would have been confident that the pairwise comparisons were consistent and that the results could be reliably used for further analysis and decision-making. By carrying out this consistency check, we can validate the accuracy of the expert judgments involved in the AHP process, ensuring that the final weightings used to prioritize climate change mitigation strategies are dependable and meaningful. If inconsistencies persist, further refinement of the comparisons would be necessary to improve the reliability of the decision-making process. This step is crucial in maintaining the integrity of the AHP results and ensuring that the chosen strategies reflect well-considered and logical expert opinions.

In this case, since $CR < 0.10$, we conclude that the judgments made during the pairwise comparisons are consistent, confirming the reliability of our evaluation of the criteria for climate change mitigation strategies in coastal regions.

To evaluate the effectiveness of various climate change mitigation strategies for coastal regions, we use the pairwise comparison method for each strategy under the four criteria: Environmental Impact, Cost-Effectiveness, Social Acceptance, and Long-Term Sustainability. This systematic approach enables us to derive the relative importance of each strategy based on the different criteria. We construct pairwise comparison matrices for each criterion, allowing for a structured, comparative assessment that ultimately guides decision-making. Below is a detailed explanation of how each criterion and its respective strategies are evaluated:

Table 4
The calculation steps for Consistency Ratio.

Calculation step	Formula/description	Result
New Vector V	Multiply the original matrix by the weights vector	[7.56, 4.07, 1.37, 3.61]
Calculate λ_{max}	Average of elements in V	3.77
Calculate CI	CI = (3.77–4) / (4–1)	−0.07
Calculate CR	CR = − 0.07 / 0.9	−0.078

Table 3
Normalized matrix and weights.

Criteria	Environmental impact	Cost-effectiveness	Social acceptance	Long-term sustainability	Weight
Environmental impact	0.52	0.60	0.59	0.53	0.56
Cost-effectiveness	0.17	0.20	0.35	0.26	0.24
Social acceptance	0.10	0.07	0.12	0.07	0.09
Long-term sustainability	0.13	0.13	0.24	0.13	0.16

3.1. Environmental Impact criterion

The pairwise comparison matrix for Environmental Impact (Table 5) reflects expert judgments about how each strategy compares to the others regarding its ecological benefits. For instance, Mangrove Restoration is rated the most important, with a score of 9 compared to Relocation Programs, indicating its higher environmental value in mitigating climate change. Similarly, Mangrove Restoration is also viewed as significantly more beneficial than Zoning & Codes and Seawalls, with scores of 7 and 5, respectively. These values are based on the belief that nature-based solutions like mangrove restoration provide critical ecosystem services, such as carbon sequestration and coastal protection. After constructing this matrix, we normalize the values by dividing each entry by the sum of its column. The resulting normalized matrix (not shown here) reflects the proportional importance of each strategy relative to others under the Environmental Impact criterion. The final weights for each strategy are then obtained by averaging the normalized values across each row. These weights measure how much each strategy contributes to the overall environmental benefit.

3.2. Cost-Effectiveness Criterion

In Table 6, the pairwise comparison matrix for Cost-Effectiveness assesses how each strategy compares regarding financial feasibility and efficiency. Mangrove Restoration is again ranked highly, receiving the highest scores compared to Zoning and codes and Relocation Programs. However, Relocation Programs receive lower scores due to the high costs associated with displacement, which explains their relatively lower cost-effectiveness ranking. Following the normalization process, we obtain the proportional importance of each strategy regarding cost, ensuring that the most economically viable strategies are prioritized in the final analysis.

3.3. Social Acceptance Criterion

Table 7 illustrates the pairwise comparison for Social Acceptance, focusing on how each strategy aligns with community needs and social feasibility. Mangrove Restoration ranks highest, reflecting the growing recognition of community-based ecological restoration efforts as effective and culturally resonant. In contrast, strategies like Seawalls and Coral Reef Protection receive lower scores, as they are often seen as less socially acceptable due to their impacts on local communities and economies. After normalizing the matrix, we calculate the weights, which reflect the public acceptance of each strategy, guiding policymakers to choose approaches that benefit the environment and align with the needs of coastal populations.

3.4. Long-Term Sustainability Criterion

Finally, Table 8 presents the pairwise comparison matrix for Long-Term Sustainability, emphasizing each strategy's long-term viability and adaptability. Mangrove Restoration scores highly here, reflecting its enduring ecological and socio-economic benefits, such as carbon sequestration and natural coastal protection. In comparison, strategies like Relocation Programs rank lower, as the social costs and the ongoing need for adaptation to a changing climate compromise their long-term sustainability. After normalization, the resulting weights show the relative long-term feasibility of each strategy in coastal climate change mitigation.

After normalizing the final matrices for each criterion—Environmental Impact, Cost-Effectiveness, Social Acceptance, and Long-Term Sustainability—and calculating the average values for each strategy, we can derive a set of weights that quantitatively represent the effectiveness of each strategy under these criteria. These weights are crucial for prioritizing the most appropriate climate change mitigation strategies for coastal regions, as they provide an objective basis for decision-making. This approach ensures that strategies are evaluated not only on their individual merits but also in relation to one another, based on the comprehensive goals of mitigating climate change impacts effectively. By systematically evaluating each criterion and assigning appropriate weights, we develop a structured decision framework that aids in selecting the best strategies for addressing the unique challenges posed by climate change. The next step in the AHP process involves aggregating the results from the pairwise comparison matrices and the corresponding weights for each criterion. This aggregation is performed by multiplying the weight assigned to each criterion by the normalized weight assigned to each strategy under that criterion. Once this is done for all criteria, the results are summed to calculate a total score for each strategy. These final scores reflect how well each strategy aligns with the overall goals of environmental sustainability, economic feasibility, social acceptance, and long-term resilience. As shown in Table 9, the strategy of Mangrove Restoration receives the highest total score of 0.65, reflecting its strong alignment with the four key criteria. This is because Mangrove Restoration scores highly in Environmental Impact (0.32) due to its significant role in protecting ecosystems and mitigating climate-related risks such as coastal erosion and flooding. It also has a favorable cost-effectiveness score (0.15), reflecting its relatively low cost compared to engineered alternatives like seawalls. Although it ranks lower in Social Acceptance (0.08), its long-term sustainability (0.10) adds to its overall effectiveness, positioning it as the most balanced and effective strategy for mitigating climate change impacts in coastal regions.

Zoning and Building Codes rank second with a total score of 0.58. This strategy performs well in cost-effectiveness (0.18) and long-term sustainability (0.12), as it offers a regulatory solution that ensures resilience to climate change impacts over time. However, it scores lower in Environmental Impact (0.22) and Social Acceptance (0.06), as it may face resistance from local communities and stakeholders who would need to comply with new regulations. Seawalls are ranked third with a total score of 0.48. While they offer significant immediate protection against coastal flooding and erosion (as indicated by their relatively high cost-effectiveness score of 0.20), they are less favorable in terms of environmental impact (0.15), as they may disrupt natural coastal ecosystems, and social acceptance (0.05), due to concerns over their long-term ecological consequences. Additionally, their relatively low long-term sustainability score (0.08) reflects their potential to require ongoing maintenance and updates to remain effective. Coral Reef Protection scores 0.46, ranking fourth. Though it has a strong environmental impact (0.18) and offers some long-term benefits (0.09), it is less cost-effective (0.12) and faces social acceptance challenges (0.07) due to the difficulty in securing long-term funding and overcoming resistance from communities that may be affected by conservation policies. Finally,

Table 5
Pairwise comparison for environmental impact.

Strategy	Mangrove restoration	Zoning & codes	Seawalls	Coral reef protection	Relocation programs
Mangrove restoration	1	3	5	7	9
Zoning & codes	1/3	1	3	5	7
Seawalls	1/5	1/3	1	3	5
Coral reef protection	1/7	1/5	1/3	1	3
Relocation programs	1/9	1/7	1/5	1/3	1

Table 6
Pairwise comparison for cost-effectiveness.

Strategy	Mangrove restoration	Zoning & codes	Seawalls	Coral reef protection	Relocation programs
Mangrove restoration	1	2	3	5	7
Zoning & codes	1/2	1	2	3	5
Seawalls	1/3	1/2	1	2	4
Coral reef protection	1/5	1/3	1/2	1	3
Relocation programs	1/7	1/5	1/4	1/3	1

Table 7
Pairwise comparison for social acceptance.

Strategy	Mangrove restoration	Zoning & codes	Seawalls	Coral reef protection	Relocation programs
Mangrove restoration	1	3	5	7	9
Zoning & codes	1/3	1	2	4	5
Seawalls	1/5	1/2	1	2	3
Coral reef protection	1/7	1/4	1/2	1	2
Relocation programs	1/9	1/5	1/3	1/2	1

Table 8
Pairwise comparison for long-term sustainability.

Strategy	Mangrove restoration	Zoning & codes	Seawalls	Coral reef protection	Relocation programs
Mangrove restoration	1	2	4	5	8
Zoning & codes	1/2	1	3	4	6
Seawalls	1/4	1/3	1	2	3
Coral reef protection	1/5	1/4	1/2	1	2
Relocation programs	1/8	1/6	1/3	1/2	1

Table 9
Aggregating results to determine prioritized strategies.

Strategy	Environmental Impact (0.56)	Cost-Effectiveness (0.24)	Social Acceptance (0.09)	Long-Term Sustainability (0.16)	Total Score
Mangrove restoration	0.32	0.15	0.08	0.10	0.65
Zoning & codes	0.22	0.18	0.06	0.12	0.58
Seawalls	0.15	0.20	0.05	0.08	0.48
Coral reef protection	0.18	0.12	0.07	0.09	0.46
Relocation programs	0.11	0.09	0.12	0.06	0.38

Relocation Programs rank lowest with a total score of 0.38. While they perform well in Social Acceptance (0.12), reflecting the importance of community involvement and the potential to mitigate the effects of extreme weather events, their environmental impact (0.11) and cost-effectiveness (0.09) are less favorable. The strategy's long-term sustainability (0.06) is also a concern, as it involves significant displacement and social disruption. This aggregation process provides a comprehensive ranking of the strategies, considering all relevant criteria. The total scores clearly illustrate each strategy's relative importance and effectiveness in addressing climate change mitigation challenges in coastal regions. The structured approach ensures that decision-makers can prioritize the strategies that offer the best balance of environmental, social, and economic benefits, while also guaranteeing long-term resilience to climate impacts.

The final aggregated scores provide a comprehensive overview of each strategy's relative effectiveness in climate change mitigation in coastal regions. **Mangrove Restoration** emerges as the highest-scoring strategy with a total score of **0.65**, indicating its significant potential in terms of environmental impact, cost-effectiveness, social acceptance, and long-term sustainability. **Zoning & Codes** follows closely with a score of **0.58**, suggesting it is also viable. In contrast, **Seawalls**, **Coral Reef Protection**, and **Relocation Programs** score lower, indicating they may not be as effective as the top two strategies in the context of the established criteria. This structured approach facilitates informed decision-making by highlighting the strategies that best align with environmental goals while considering economic and social factors, ultimately supporting sustainable development in coastal areas facing climate change challenges.

Fig. 2 illustrates the Analytic Hierarchy Process (AHP) framework used in this study to systematically prioritize climate change mitigation strategies for coastal regions. The diagram starts with the goal at the top: identifying and prioritizing the most effective mitigation strategies. Beneath the goal are the four main criteria—Environmental Impact, Cost-Effectiveness, Social Acceptance, and Long-Term Sustainability—-that guide the evaluation process. These criteria serve as the foundation for comparing the different strategies, ensuring a comprehensive assessment that balances ecological, economic, and social considerations. The strategies being assessed, including Mangrove Restoration, Zoning & Building Codes, Seawalls, Coral Reef Protection, and Relocation Programs, are positioned at the bottom level of the hierarchy. Each of these strategies is evaluated based on how well they perform under each of the four criteria, with a pairwise comparison approach used to determine their relative importance. The AHP framework uses a weighted scoring system, where each criterion is assigned, a weight reflecting its significance in the overall decision-making process, and each strategy is scored against these criteria. The final scores are aggregated to determine the prioritized strategies, offering a clear recommendation for the most balanced, effective, and feasible climate change mitigation measures. This systematic evaluation not only provides transparency in the decision-making process but also ensures that the selected strategies align with the region's long-term environmental, economic, and social needs, fostering greater resilience to climate change.

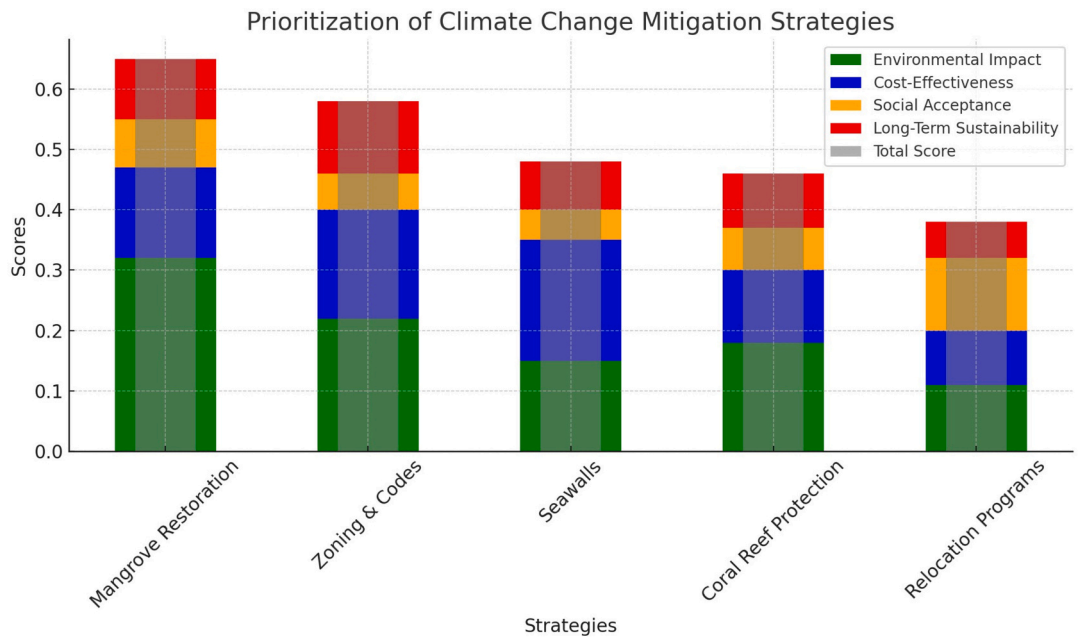

Fig. 2. Analytical Hierarchy Process Framework for Prioritizing Climate Change Mitigation Strategies for Coastal Regions.

4. Discussion

The prioritized mitigation strategies from this study offer valuable insights for policymakers in coastal climate management. The results indicate a strong preference for ecosystem-based solutions, such as Mangrove Restoration, ranked highest across criteria, particularly for environmental impact and sustainability. This emphasis on ecosystem restoration aligns well with a shift in coastal management policies prioritizing nature-based solutions, which protect coastlines, enhance biodiversity, and support local communities. However, some existing policies still heavily favor engineering solutions, like Seawalls, for their immediate protective benefits despite these being ranked lower in long-term sustainability in our analysis. Policymakers might consider adjusting funding allocations to balance immediate and sustainable solutions, promoting regulatory approaches like Zoning and Building Codes, which scored second overall and offer a cost-effective means to build resilience over time. The findings emphasize that strategies with community and ecosystem benefits, such as Mangrove Restoration and Coral Reef Protection, should be prioritized in national and local climate action plans. Local governments in highly vulnerable coastal areas could invest in these strategies, delivering ecological and socio-economic benefits. By prioritizing these approaches, policymakers would address not only environmental impacts but also the socio-economic needs of coastal communities. When comparing these findings with previous studies, a consistent theme emerges ecosystem-based strategies are frequently highlighted as high-priority options for coastal resilience. Previous research has shown that Mangrove Restoration and Coral Reef Protection provide multiple benefits, including carbon sequestration, storm buffering, and habitat conservation. This aligns with our findings on their high environmental impact and sustainability rank. For example, a study by Narayan et al. (2016) also ranked mangrove ecosystems highly for reducing wave energy and protecting shorelines, which is consistent with our findings (Narayan et al., 2016). This indicates a broader trend in coastal management toward integrating ecological restoration into climate resilience frameworks. However, certain discrepancies exist. Some past studies prioritized Engineering Solutions such as seawalls more highly, particularly in densely populated urban coastal areas where immediate protection is critical due to high economic value and population density. These discrepancies can be attributed to differences in geographic focus and chosen criteria. Studies focusing on urban coastal areas prioritize cost-effectiveness and immediate protective capacity. In contrast, this study takes a holistic view that values long-term sustainability and social acceptance, which are critical in rural or ecologically sensitive coastal regions.

This study employs the Analytic Hierarchy Process (AHP) to prioritize climate change mitigation strategies for coastal regions. Still, alternative approaches, such as PROMETHEE, TOPSIS, or the Delphi method, are also feasible. PROMETHEE offers strong ranking consistency and is effective for complex datasets, while TOPSIS emphasizes proximity to ideal solutions, making it suitable for scenarios with measurable quantitative data. The Delphi method facilitates expert consensus and can be particularly valuable in addressing subjective decision-making contexts. However, AHP was chosen for its ability to decompose complex problems into a hierarchical structure, enabling detailed pairwise comparisons of ecological, social, and economic criteria. This method is advantageous in coastal contexts, where criteria often conflict, such as balancing cost-effectiveness with long-term sustainability. Previous studies, like Arda et al. (2024), have demonstrated that while BWM may require fewer comparisons than AHP, its simplicity limits nuanced assessments across multiple interdependent criteria. Similarly, Bagheri et al. (2021) applied Delphi-AHP in coastal vulnerability studies, underscoring AHP's strength in integrating expert opinions with quantitative data to produce actionable rankings. In this study, AHP enabled the identification of Mangrove Restoration as the most effective strategy due to its environmental and socio-economic benefits while also reflecting the trade-offs associated with other strategies like Seawalls or Relocation Programs. The hierarchical structure provided by AHP ensures that results comprehensively address the unique complexities of coastal regions, a challenge less effectively managed by other methods that may lack the capacity to integrate diverse, qualitative inputs. This methodological choice strengthens the study's relevance, as it aligns prioritization with both ecological and human needs, producing robust, actionable insights for policymakers.

Numerous studies have explored the interplay between climate change, water resources, and agricultural management, using various methodologies to identify optimal adaptation strategies. One study in Iran's Gharanghu Basin assessed the water-climate-agriculture nexus by evaluating five adaptive scenarios to address reduced water resources under climate change. These scenarios included altering cropping patterns, reducing cultivation areas, reusing agricultural wastewater, and

combinations thereof. Using the time reliability index, results showed that implementing a combined strategy of reducing cultivated areas and reusing wastewater increased system reliability to 85 %, demonstrating the importance of balancing water demand and supply for sustainable reservoir management(Golfam et al., 2021). Another study applied AHP and TOPSIS to determine the best agricultural water management scenario under future climate conditions. AHP ranked a 25 % reduction in water demand as the optimal solution, while TOPSIS identified a 15 % reduction as the best scenario. This discrepancy highlights the value of addressing uncertainty in MCDM methods to ensure robust decisions under changing conditions(Golfam et al., 2019b). Similarly, in East Azerbaijan, Iran, reservoir operation under climate-change projections was optimized using VIKOR and FOWA methods, identifying effective water-supply strategies to sustain agriculture and cropping patterns (Golfam et al., 2019a). In another application, the HadCM3 climate model predicted increased irrigation demand and reduced inflows, prompting recommendations to delay planting dates, apply water stress, and improve irrigation efficiency. Using MO-GP, the study found that enhanced irrigation efficiency could maintain crop yields while reducing water supply vulnerability(Ashofteh et al., 2017). Additionally, a separate study focused on the Gharanghu irrigation network revealed that applying climate adaptation strategies, such as demand reductions of 5–20 %, could increase system flexibility by up to 38 % and reduce vulnerability by 28 % (Ashofteh et al., 2019). Compared to the findings in this paper, where the AHP prioritized Mangrove Restoration and other mitigation strategies based on multi-criteria evaluation, these studies underline the critical role of MCDM approaches in addressing region-specific water and agricultural challenges, effectively balancing environmental and socio-economic needs.

4.1. Limitations of the study

This study provides a structured approach to prioritize climate mitigation strategies for coastal regions; however, several limitations should be acknowledged. First, while comprehensive, the scope of selected criteria may not capture every aspect relevant to all coastal communities. For instance, additional criteria such as ecosystem connectivity or cultural significance could enhance the accuracy of prioritization, especially in regions where these factors play an essential role. Expanding criteria to cover broader socio-cultural aspects might provide a more complete representation of stakeholder priorities. Another limitation lies in data availability. The reliance on past literature to determine pairwise comparisons introduces a potential bias, as the data may not represent the latest conditions or regional variations. Future studies could benefit from more localized data, reflecting region-specific conditions and priorities. Furthermore, while the survey minimizes reliance on expert judgment by drawing from existing literature, some comparisons inevitably involve subjective interpretations. Incorporating broader stakeholder input, including perspectives from local communities, NGOs, and businesses, could add depth to the analysis, making it more inclusive of diverse viewpoints. Lastly, this analysis is static, reflecting current conditions and criteria weights. However, climate impacts and coastal dynamics are constantly evolving. Future research could employ dynamic models or real-time data integration, allowing for a prioritization model that adapts to environmental, economic, and social changes over time. Employing such approaches could ensure that mitigation strategies remain relevant and responsive to shifting coastal climate risks and community needs.

4.2. Recommendations and future research direction

Several actionable recommendations can be made for policymakers, urban planners, and local communities to leverage this study's findings effectively. First, promoting and funding ecosystem-based strategies is essential. Mangrove Restoration should be prioritized with appropriate policy support and funding, given its top ranking. This strategy offers combined ecological and socio-economic benefits, making it well-suited for regions requiring long-term, sustainable solutions. Next, it is crucial to implement zoning and regulatory controls. With a high ranking and a score of 0.58 for cost-effectiveness and sustainability, Zoning and Building Codes are vital for establishing resilient land-use practices. Local governments can introduce zoning restrictions in vulnerable areas while incentivizing building practices that account for climate risks, contributing to coastal resilience. Additionally, it is recommended that engineering solutions be adopted using hybrid approaches. Seawalls, which scored 0.48, are particularly effective in high-risk, densely populated areas. Policymakers should consider pairing seawalls with natural barriers, such as mangrove buffers, to enhance their protective capabilities and long-term sustainability. Lastly, it is essential to plan for managed retreats where necessary. While Relocation Programs ranked lowest with a score of 0.38, they remain essential for communities facing extreme risks. Proactively planning for managed retreat can mitigate the socio-economic impacts on these communities, making it a vital strategy for high-risk coastal areas.

Future research could further extend the value of this study by expanding the AHP framework to other geographic contexts. Applying the AHP model to diverse coastal settings, including island nations or polar regions, would test the generalizability of these results and reveal additional regional priorities. Moreover, incorporating broader socio-economic and cultural criteria would enhance the analysis. By adding factors such as cultural significance, equity impacts, or tourism considerations, the AHP framework could become more comprehensive, reflecting coastal zones' multifaceted socio-economic roles. Finally, using complementary decision-making models could enrich the prioritization process. Pairing AHP with frameworks like the Delphi method or multi-criteria decision analysis (MCDA) may provide valuable insights, especially when integrating dynamic, real-time data sources for greater adaptability. These recommendations and future research directions collectively offer a roadmap for policymakers seeking to balance immediate coastal protection with sustainable, long-term climate resilience. Emphasizing strategies such as ecosystem restoration and regulatory approaches aligns coastal management with community needs and environmental priorities, ultimately fostering a resilient foundation for coastal regions in the face of climate change.

5. Conclusions

This study provides a data-driven framework for prioritizing climate change mitigation strategies for coastal regions using the Analytic Hierarchy Process (AHP). The findings underscore the effectiveness of nature-based solutions, particularly Mangrove Restoration, which emerged as the top strategy due to its significant environmental impact, long-term sustainability, and moderate social acceptance. Other strategies, such as Zoning and Building Codes, Seawalls, and Coral Reef Protection, followed in the ranking, with each strategy offering unique advantages but presenting limitations in areas such as cost-effectiveness and social acceptance. Finally, Relocation Programs, while essential for communities facing extreme climate impacts, ranked the lowest due to their resource-intensive nature and associated social challenges. These results emphasize the need for a balanced approach incorporating nature-based solutions and regulatory frameworks, which can collectively enhance coastal resilience while addressing the socio-economic needs of local communities.

While this study provides valuable insights, there are several limitations and areas for future improvement. First, the assessment is based on expert opinions and pairwise comparisons, which, while rigorous, can introduce subjectivity. Incorporating broader stakeholder perspectives, including those from local communities and governments, could improve the robustness and inclusivity of the evaluation. Second, the study primarily focuses on ecological, social, and economic criteria. Still, it may benefit from integrating additional dimensions, such as climate risk and ecosystem service valuation, to offer a more

comprehensive assessment. Third, the study uses a fixed set of strategies for coastal adaptation; however, emerging strategies—such as floating structures or green infrastructure—could be evaluated in future studies to ensure that all innovative approaches are considered.

For future research, a dynamic, adaptive model could be developed that accounts for the changing nature of climate impacts, allowing for adjustments over time as new data becomes available. Moreover, the framework could be applied to specific case studies or regional assessments to tailor the strategies to different coastal environments and socio-economic contexts. Sensitivity analysis could also be expanded to explore the robustness of the results under varying assumptions or uncertainties related to climate change projections.

In summary, while the current study provides a strong foundation for prioritizing climate change mitigation strategies in coastal regions, the incorporation of a broader range of perspectives, additional criteria, and emerging strategies will strengthen the decision-making process and provide more targeted solutions for enhancing coastal resilience in the face of climate challenges.

CRediT authorship contribution statement

Jie Liu: Writing – original draft, Formal analysis, Data curation. **Xiaohui Liu:** Writing – original draft, Methodology, Formal analysis. **Ancheng Zhu:** Writing – original draft, Methodology, Investigation. **Xitao Wang:** Writing – original draft, Visualization, Resources, Project administration. **Qianqian Yu:** Writing – original draft, Methodology, Investigation. **Lu Chen:** Writing – original draft, Visualization, Data curation, Conceptualization, Supervision. **Tariq J. Al-Musawi:** Formal analysis, Investigation, Writing – original draft. **Mahnaz Aasal:** Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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